

GearDesign Report

Cylindrical gear(pair)

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1. Summary

1.1. Gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.0000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	21.0400	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	29.0000	65.0000
Facewidth (mm)	b	35.0000	40.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	100.7150	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Reference diameter (mm)	d	62.1430	139.2860
Tip diameter (mm)	d_a	66.143 (66.143 / 66.043)	143.286 (143.286 / 143.186)
Root diameter (mm)	d_f	57.143 (56.951 / 56.841)	134.286 (134.025 / 133.888)
Number of teeth spanned	k	4.0000	9.0000
Base tangent length (no backlash) (mm)	W_k	21.654 (21.588 / 21.550)	52.402 (52.313 / 52.266)
Effective Diameter of ball/pin (mm)	D_M	3.5000	3.5000
Diametral measurement over two balls (mm)	M_{dk}	67.042 (66.873 / 66.775)	144.292 (144.047 / 143.917)

1.2. Total safety factor

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]	Required S.F
Contact S.F	1.2134		1.2438		1.0000
Bending S.F	2.2745		2.6241		1.4000
Scuffing S.F (Flash Temp.)	3.2979				2.0000
Micropitting S.F	1.2419				1.0000

1.3. Total damage & life

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]
Contact Damage [%]	0.2000	0.0000	0.1000	0.0000
Bending Damage [%]	0.0000	0.0000	0.0000	0.0000
Contact Life [hr]	∞	∞	∞	∞
Bending Life [hr]	∞	∞	∞	∞
Total Req. Life [hr]	10000.0000			

1.4. Others

	Gear1 - Gear2
Transverse Contact Ratio	1.4880
Overlap Ratio	1.9999
Total Contact Ratio	3.4879
Worst Pitch Line Velocity [m/s]	32.5380
Total Efficiency [%]	98.5094
Worst Mesh Loss [W]	542.0281
Worst Churning Loss [W]	1693.8301
Worst Peak-to-Peak T.E. [μm]	0.2615
Gear Total Mass [kg]	1.5219
Hunting tooth	Complete

1.5. Summary - Load cases

[illegible]

2. Detail geometry

2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.0000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	21.0400	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	29.0000	65.0000
Facewidth (mm)	b	35.0000	40.0000
Profile shift coefficient	x	0.0000	0.0000
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	100.7146	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Addendum coefficient	h^*_aP	1.0000	1.0000
Dedendum coefficient	h^*_fP	1.2500	1.2500
Root radius coefficient	ρ^*_fP	0.3800	0.3800
Tip form height coefficient	h^*_{FaP}	0.9500	0.9500
Ramp angle (deg)	α_{KP}	50.0000	50.0000
Protuberance height coefficient	h^*_{prP}	0.0000	0.0000
Protuberance angle (deg)	α_{prP}	0.0000	0.0000
Tip alteration (mm)	k_{m_n}	0.0000	0.0000
Tip radius coefficient (for topping)	ρ^*_{aP}	0.0000	0.0000
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Young's modulus (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Poisson's ratio	ν	0.3000	0.3000
Density (kg/m ³)	ρ	7830.0000	7830.0000
Coeff. of thermal expansion (10 ⁻⁸ /°C)	α_{coeff}	11.5000	11.5000
Surface hardness	HRC	61.0000	61.0000
Core hardness	HBW	325.0000	325.0000
Tensile strength (N/mm ²)	σ_B	1200.0000	1200.0000
Yield strength (N/mm ²)	σ_s	850.0000	850.0000
Specific heat capacity (J/kg.K)	c_M	485.0000	485.0000
Specific heat conductivity (W/m.K)	λ_M	50.0000	50.0000
Mean roughness height, root (um)	R_{ZF}	20.0000	20.0000
Mean roughness height, flank (um)	R_{ZH}	4.8000	4.8000
Roughness average value, root (um)	R_{AF}	3.0000	3.0000
Roughness average value, flank (um)	R_{AH}	0.6000	0.6000

	Symbol	Pair1 Gear1	Pair1 Gear2
Fatigue strength for bending stress (N/mm ²)	σ_{Flim}	430.0000	430.0000
Fatigue strength for contact stress (N/mm ²)	σ_{Hlim}	1500.0000	1500.0000

2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Gear ratio	GR	2.2414	
Transverse module (mm)	m_t	2.1429	
Working normal pressure angle (deg)	α_{wn}	20.0000	
Transverse pressure angle (deg)	α_t	21.3043	
Working transverse pressure angle (deg)	α_{wt}	21.3043	
Working transverse pressure angle, e/i (deg)	$\alpha_{wt}, e/i$	21.3203 / 21.2882	
Base helix angle (deg)	β_b	19.7166	
Helix angle at operating pitch circle (deg)	β_w	21.0400	
Radial backlash without tolerance (mm)	j_{r0}	0.0000	
Cumferential backlash without tolerance (mm)	j_{t0}	0.0000	
Min/max circumferential backlash by flank tolerance for each gear (mm)	$j_t, e/i$	0.0750 / 0.1179	0.1018 / 0.1554
Min/max normal backlash by flank tolerance for each gear (mm)	$j_n, e/i$	0.0658 / 0.1034	0.0893 / 0.1363
Cumferential backlash, e/i (mm)	$j_{tw}, e/i$	0.1682 / 0.2818	
Normal backlash, e/i (mm)	$j_{nw}, e/i$	0.1475 / 0.2471	
Angluar backlash, min (deg)	j_θ, min	0.3102	0.1384
Angluar backlash, max (deg)	j_θ, max	0.5196	0.2318
Effective facewidth (mm)	b_{eff}	35.0000	
Sum of the profile shift coefficients	Sum_x	-0.0000	
Generating profile shift coefficient	$x_E, e/i$	-0.0481 / -0.0756	-0.0653 / -0.0996
Prevent undercut profile shift coefficient	x_{Emin}	-1.0508	-3.5965
Max. profile shift coeff. (min top land)	x_{Emax}	1.4834	2.5029
Reference center distance, $x_1=x_2=0$ (mm)	a_d	100.7146	
Working center distance (mm)	a_w	100.7146	
Reference diameter (mm)	d	62.1431	139.2862
Operating pitch diameter (mm)	d_w	62.1431	139.2862
Operating pitch diameter with tolerance (mm)	$d_w, e/i$	62.1499 / 62.1363	139.3014 / 139.2710
27		66.1431	143.2862
Tip diameter with tolerance (mm)	$d_a, e/i$	66.1431 / 66.0431	143.2862 / 143.1862

	Symbol	Pair1 Gear1	Pair1 Gear2
Root diameter (mm)	d _f	57.1431	134.2862
Root diameter with tolerance (mm)	d _f , e/i	56.9508 / 56.8409	134.0252 / 133.8878
Tip form diameter (mm)	d _{Fa}	65.9431	143.0862
Tip form diameter with tolerance (mm)	d _{Fa} , e/i	65.7319 / 65.6117	142.8127 / 142.6691
Root form diameter (mm)	d _{Ff}	59.0410	135.6746
Root form diameter with tolerance (mm)	d _{Ff} , e/i	58.9395 / 58.8836	135.4667 / 135.3586
Active tip diameter (EAP) (mm)	d _{Na}	65.9431	143.0862
Active tip diameter with tolerance (mm)	d _{Na} , e/i	65.7319 / 65.6117	142.8127 / 142.6691
Active root diameter (SAP) (mm)	d _{Nf}	59.3165	136.2778
Active root diameter with tolerance (mm)	d _{Nf} , e/i	59.5554 / 59.4479	136.5114 / 136.3952
Mean diameter for mass calculation (mm)	d _m	61.4944	138.5964
Gear mass (kg)	m	0.3877	1.1342
Gear inertia (kg.m ²)	J	0.0000	0.0000
Nominal tip clearance (mm)	c	0.5000	0.5000
Effective tip clearance, e/i (mm)	c, e/i	0.7602 / 0.6195	0.7121 / 0.5852
Min. margin of involute contact, (d _{Nf_i} - d _{Ff_e}) / 2 (mm)	c _{Nff}	0.2542	0.4643

2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Tooth depth (mm)	h	4.5000	4.5000
Tooth depth, e/i (mm)	h, e/i	4.6511 / 4.5462	4.6992 / 4.5805
Working depth (mm)	h _w	4.0000	
Addendum (mm)	h _a	2.0000	2.0000
Addendum, e/i (mm)	h _a , e/i	2.0000 / 1.9500	2.0000 / 1.9500
Dedendum (mm)	h _f	2.5000	2.5000
Dedendum, e/i (mm)	h _f , e/i	2.5962 / 2.6511	2.6305 / 2.6992
Number of virtual gear teeth	z _n	35.0622	78.5877
Tooth thickness coeff.	s _{nc}	1.5708	1.5708
Transverse tooth thickness (mm)	s _t	3.3660	3.3660
Normal tooth thickness (mm)	s _n	3.1416	3.1416
Normal tooth thickness at d _a (mm)	s _{an}	1.5049	1.5981
Normal tooth thickness at d _a , e/i (mm)	s _{an} , e/i	1.4801 / 1.3888	1.5432 / 1.4494
Normal tooth thickness at d _{Fa} (mm)	s _{Fan}	1.6025	1.6828
Normal tooth thickness at d _{Fa} , e/i (mm)	s _{Fan} , e/i	1.6874 / 1.5883	1.7601 / 1.6493
Normal space width at d _f (mm)	e _{fn}	0.0000	1.5182
Normal space width at d _f , e/i (mm)	e _{fn} , e/i	0.0000 / 0.0000	1.5433 / 1.5572

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal pitch (mm)	p_n	6.2832	
Transverse pitch (mm)	p_t	6.7320	
Normal base pitch (mm)	p_bn	5.9043	
Transverse base pitch (mm)	p_bt	6.2720	
Axial pitch (mm)	p_x	17.5010	
Lead (mm)	p_z	507.5276	1137.5619

2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Length of the path of contact (A to E) (mm)	g_α	9.3330	
Length of path of contact, e/i (mm)	g_α, e/i	8.8154 / 8.4551	
Length of approach POC of pinion (A to C) (mm)	g_f1	4.8383	
Length of approach POC of pinion, e/i (mm)	g_f1, e/i	4.5332 / 4.4913	
Length of recess POC of pinion (C to E) (mm)	g_a1	4.4947	
Length of recess POC of pinion, e/i (mm)	g_a1, e/i	4.2823 / 4.2636	
Length T1 - A (mm)	T1A	6.4506	
T1A tolerance (mm)	e/m/i	6.9794 / 6.8629 / 6.7464	
Length T1 - B (mm)	T1B	9.5116	
T1B tolerance (mm)	e/m/i	9.2899 / 9.2262 / 9.1625	
Length T1 - C (mm)	T1C	11.2889	
T1C tolerance (mm)	e/m/i	11.2983 / 11.2889 / 11.2796	
Length T1 - D (mm)	T1D	12.7226	
T1D tolerance (mm)	e/m/i	13.2513 / 13.1349 / 13.0184	
Length T1 - E (mm)	T1E	15.7836	
T1E tolerance (mm)	e/m/i	15.5619 / 15.4982 / 15.4345	
Length T2 - A (mm)	T2A	30.1411	
T2A tolerance (mm)	e/m/i	29.8150 / 29.7288 / 29.6426	
Length T2 - B (mm)	T2B	27.0801	
T2B tolerance (mm)	e/m/i	27.4595 / 27.3655 / 27.2715	
Length T2 - C (mm)	T2C	25.3028	
T2C tolerance (mm)	e/m/i	25.3237 / 25.3028 / 25.2818	
Length T2 - D (mm)	T2D	23.8691	
T2D tolerance (mm)	e/m/i	23.5430 / 23.4568 / 23.3706	
Length T2 - E (mm)	T2E	20.8081	
T2E tolerance (mm)	e/m/i	21.1875 / 21.0935 / 20.9996	
Length T1 - T2 (mm)	T1T2	36.5917	

	Symbol	Pair1 Gear1	Pair1 Gear2
T1T2 tolerance (mm)	e/m/i	36.6220 / 36.5917 / 36.5614	
Diameter of single contact point A (mm)	d_A	59.3165	143.0862
Diameter of single contact point B (mm)	d_B	60.9417	140.6167
Diameter of single contact point C (mm)	d_C	62.1431	139.2862
Diameter of single contact point D (mm)	d_D	63.2413	138.2703
Diameter of single contact point E (mm)	d_E	65.9431	136.2778
Transverse contact ratio	ϵ_α	1.4880	
Overlap ratio	ϵ_β	1.9999	
Total contact ratio	ϵ_γ	3.4879	
Total contact ratio with tolerance	e/m/i	3.405 / 3.377 / 3.348	
Specific sliding at Tip	ζ_a	0.4118	0.5203
Specific sliding at Root	ζ_f	-1.0847	-0.7002

2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Quality Standard	-	ISO 1328: 2013	ISO 1328: 2013
Single pitch deviation (μm)	f_pt	8.5000	8.5000
Base circle pitch deviation (μm)	f_pb	7.9000	7.9000
Sector pitch deviation over k/8 pitches (μm)	F_pk/8T	18.0000	18.0000
Profile form deviation (μm)	f_fa	8.5000	8.5000
Profile slope deviation (μm)	f_Ha	7.0000	7.0000
Total profile deviation (μm)	F_α	11.0000	11.0000
Helix form deviation (μm)	f_fb	10.0000	11.0000
Helix slope deviation (μm)	f_Hβ	9.0000	9.5000
Total helix deviation (μm)	F_β	13.0000	15.0000
Cumulative pith deviation, total (μm)	F_p	25.0000	29.0000
Runout (μm)	F_r	22.0000	26.0000
Single flank composite, tooth-to-tooth (μm)	f_isT	8.0000	8.0000
Single flank composite, total (μm)	F_isT	33.0000	37.0000
Tooth-to-tooth radial composite deviation (μm)	f_idT	9.5000	9.5000
Total radial composite deviation (μm)	F_idT	31.0000	37.0000

2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Number of teeth spanned	k	4.0000	9.0000
Base tangent length (mm)	W_k	21.6536	52.4022
Base tangent length with tolerance (mm)	W_k, e/i	21.5878 / 21.5502	52.3129 / 52.2659
Tolerance of W_k (mm)	ΔW_k , e/i	-0.0658 / -0.1034	-0.0893 / -0.1363
Diameter of contact point (mm)	d_M	61.3537	138.7902
Theoretical diameter of ball/pin (mm)	D_M.theory	3.3987	3.3672
Effective Diameter of ball/pin (mm)	D_M.eff	3.5000	3.5000
Radial single ball measurement (mm)	M_rk	33.5676	72.1667
Radial single ball measurement with tolerance (mm)	M_rk, e/i	33.4829 / 33.4339	72.0441 / 71.9791
Diametral two ball measurement (mm)	M_dk	67.0419	144.2923
Diametral two ball measurement with tolerance (mm)	M_dk, e/i	66.8727 / 66.7750	144.0471 / 143.9171
Tolerance of M_dk (mm)	ΔM_{dk}	-0.1692 / -0.2669	-0.2452 / -0.3752

3. Detail rating

3.1. General

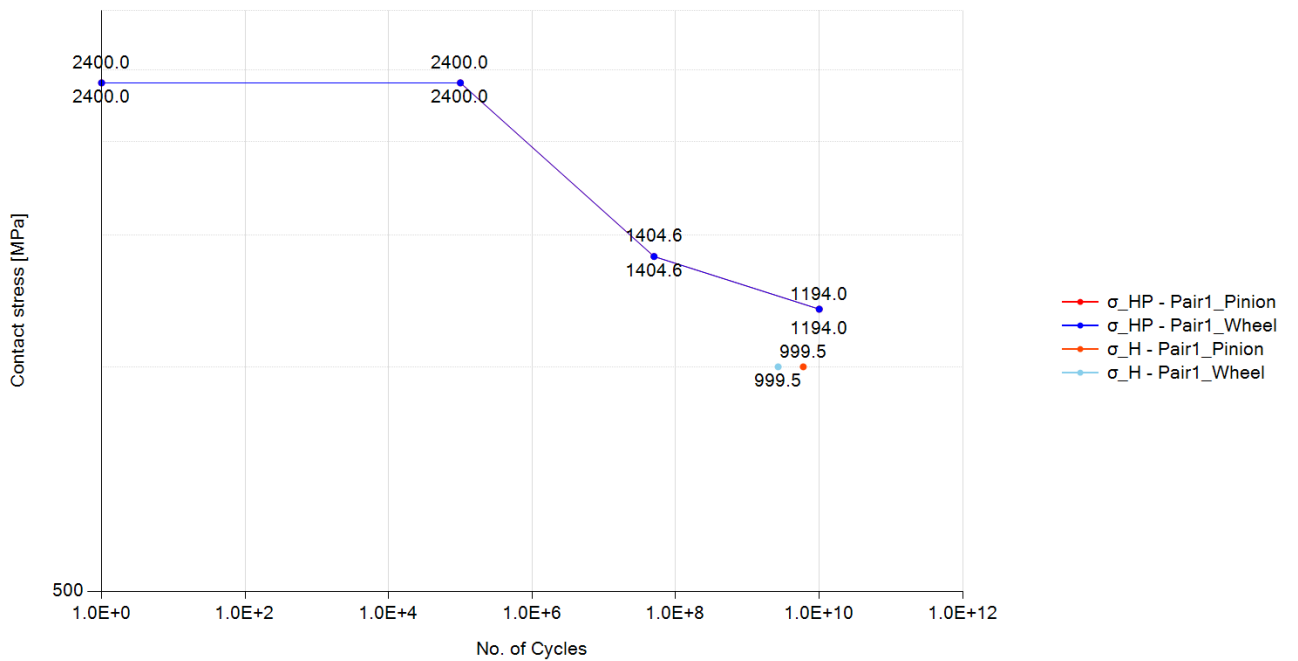
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rotational speed (rpm)	ω	10000.0000	4461.5380
Nominal tangential load at d (N)	F_t	4609.9890	
Nominal axial load at d (N)	F_a	1773.3030	
Nominal radial load at d (N)	F_r	1797.7550	
Nominal normal load at d (N)	F_n	5256.2840	
Nominal tangential load per unit facewidth (N/mm)	w	131.7140	
Circumferential speed at d (m/s)	v	32.5380	
Sliding velocity at Tip dia. (m/s)	v_g	6.8070	7.3270
Nominal tangential load at d_w (N)	F_{tw}	4609.9890	
Nominal axial load at d_w (N)	F_{aw}	1773.3030	
Nominal radial load at d_w (N)	F_{rw}	1797.7550	
Circumferential speed at d_w (m/s)	v_w	32.5380	
Running-in value (um)	y_p	0.5930	
Running-in value (um)	y_f	0.6370	
Effective single pitch & profile deviation (um)	f_{pb_eff}	7.3080	
Effective single pitch & profile deviation (um)	f_{fa_eff}	7.8620	
Correction factor	C_M	0.8000	
Gear blank factor	C_R	0.8980	
Basic rack factor	C_B	0.9750	
Material factor	E/E_{st}	1.0000	
Theoretical single stiffness (N/mm/um)	c'_{th}	18.1990	
Single tooth stiffness (N/mm/um)	c'	11.8980	
Mesh stiffness (for K_v , $KH\alpha$, $KF\alpha$), (N/mm/um)	$c_{y\alpha}$	16.2530	
Mesh stiffness (for $KH\beta$, $KF\beta$), (N/mm/um)	$c_{y\beta}$	13.8150	
Mean diameter for K_v (mm)	$d_m (K_V)$	61.6430	138.7860
Moment of inertia for K_v (kg.mm ²)	$J (K_V)$	337.2700	6162.3100
Moment of gear inertia per unit facewidth (kg.mm)	J^*	9.6360	154.0580
Relative gear mass per unit facewidth (kg/mm)	m^*	0.0115	0.0366
Relative gear mass per unit facewidth (kg/mm)	m_{red}	0.0088	
Resonance running speed (rpm)	n_{E1}	14191.8570	
Resonance ratio	N	0.7050	
Resonance ratio in main resonance ratio	N_s	0.8500	
Resonance range	-	subcritical range	
Dynamic factor	K_V	1.2260	

	Symbol	Pair1 Gear1	Pair1 Gear2
Face load factor - flank	$K_{H\beta}$	1.3000	
Face load factor - root	$K_{F\beta}$	1.2570	
Transverse load factor - flank	$K_{H\alpha}$	1.1340	
Transverse load factor - root	$K_{F\alpha}$	1.1340	
Number of load cycles	N_L	6.0000×10^9	2.6770×10^9

3.2. Pitting

S-N curve (Pitting), Load case 1, Meshing pair: Gear1 - Gear2



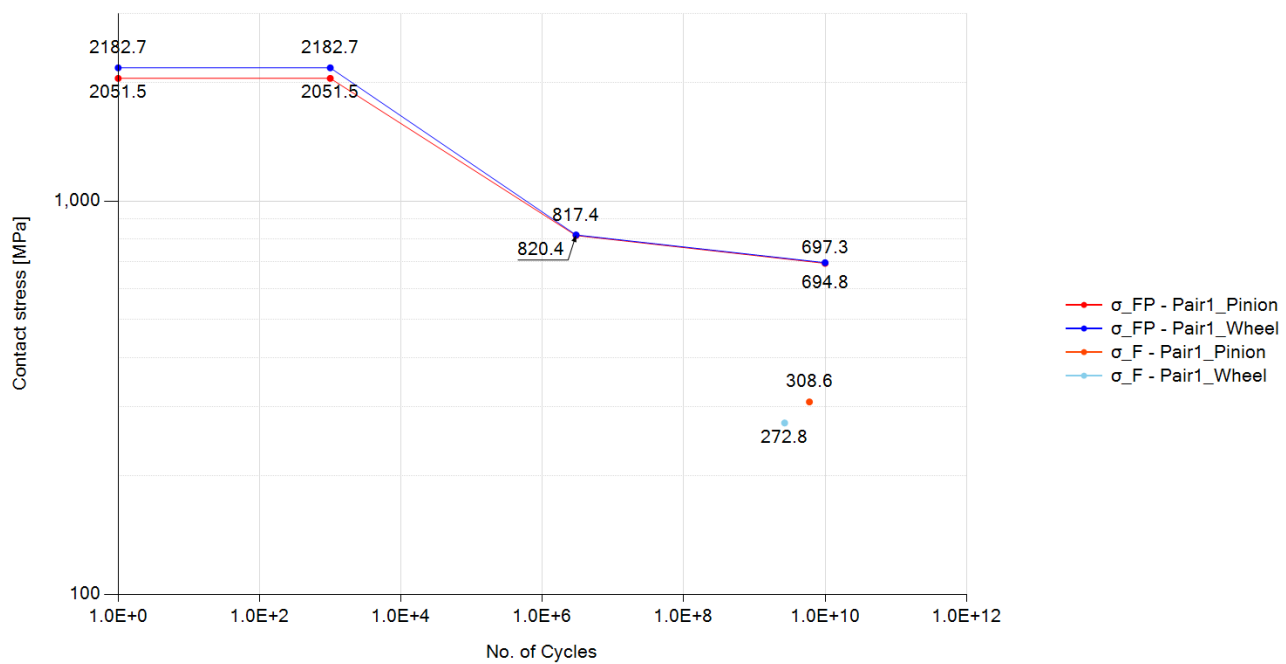
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-2:2019	
Contact ratio factor	Z_{ϵ}	0.8200	
Zone factor	Z_H	2.3580	
Elasticity factor	Z_E	189.8120	
Young's modulus for tooth flank (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Helix angle factor	Z_{β}	1.0350	
Single pair tooth contact factor (pinion)	Z_B	1.0000	
Single pair tooth contact factor (wheel)	Z_D	1.0000	
Nominal contact stress at the pitch point (N/mm ²)	σ_{H0}	665.0370	
Contact stress (N/mm ²)	σ_H	999.5060	999.5060
Life factor	Z_{NT}	0.8630	0.8850

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubricant factor	Z_L	0.9460	0.9460
Velocity factor	Z_V	1.0350	1.0350
Roughness factor	Z_R	0.9570	0.9570
Work hardening factor	Z_W	1.0000	1.0000
Size factor	Z_X	1.0000	1.0000
Permissible contact stress - static (N/mm^2)	σ_{HP_static}	2400.0000	2400.0000
Permissible contact stress - mid point (N/mm^2)	σ_{HP_mid}	0.0000	0.0000
Permissible contact stress - reference (N/mm^2)	σ_{HP_ref}	1404.6500	1404.6500
Permissible contact stress - long (N/mm^2)	σ_{HP_long}	1193.9520	1193.9520
Permissible contact stress (N/mm^2)	σ_{HP}	1212.8070	1243.2070
Safety factor for Contact	S_H	1.2130	1.2440
Contact damage (%)	-	0.1830	0.0810
Cycle to failure	-	3.2870×10^{12}	3.2870×10^{12}

3.3. Bending

S-N curve (Bending), Load case 1, Meshing pair: Gear1 - Gear2



Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-3:2019	
Tooth form factor	Y_F	1.1160	1.0510
Stress correction factor	Y_S	1.8960	2.0350

	Symbol	Pair1 Gear1	Pair1 Gear2
Working angle (deg)	α_{Fen}	18.8710	19.7970
Bending moment arm (mm)	h_{Fe}	2.0570	2.2190
Tooth thickness at root (mm)	s_{Fn}	4.1460	4.4250
Tooth root radius (mm)	ρ_F	1.1250	1.0090
Helix angle factor	Y_{β}	1.0140	
Deep tooth factor	Y_{DT}	1.0000	
Rim thickness factor	Y_B	1.0000	1.0000
Nominal tooth root stress (N/mm ²)	σ_{F0}	141.3370	124.9620
Tooth root stress (N/mm ²)	σ_F	308.5960	272.8420
Life factor	Y_{NT}	0.8590	0.8730
Relative notch sensitive factor	$Y_{\delta relT}$	0.9930	0.9970
Surface factor	Y_{RrelT}	0.9570	0.9570
Size factor	Y_X	1.0000	1.0000
Stress correction factor	Y_{ST}	2.0000	
Mean stress influence factor	Y_M	1.0000	1.0000
Technical factor	Y_T	1.0000	1.0000
Permissible tooth root stress - static (N/mm ²)	σ_{FP_static}	2051.5420	2182.7470
Permissible tooth root stress - reference (N/mm ²)	σ_{FP_ref}	817.3750	820.3540
Permissible tooth root stress - long (N/mm ²)	σ_{FP_long}	694.7690	697.3010
Permissible tooth root stress (N/mm ²)	σ_{FP}	701.9160	715.9580
Safety factor for Bending	S_F	2.2750	2.6240
Bending damage (%)	-	0.0000	0.0000
Cycle to failure	-	1.9860×10^{20}	1.1120×10^{23}

3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubrication type	-	Spray lubrication	
Lubricant temperature (deg)	θ_{oil}	80.0000	
Lubricant viscosity (mPa.s)	η_{oil}	13.1510	
Load stage according to FZG A/8, 3/90	S_{FZG}	12.0000	
Gear density (kg/m ³)	ρ_M	7830.0000	7830.0000
Gear specific heat (J/kg.K)	c_M	485.0000	485.0000
Gear heat conductivity (N/s.K)	λ_M	50.0000	50.0000
Gear roughness (μm)	R_a	0.6000	0.6000
Number of mating gears	n_p	1.0000	
Effective facewidth (mm)	b_{eff}	29.7500	

	Symbol	Pair1 Gear1	Pair1 Gear2
Relevant tip relief (μm) (Min. value = 2 μm)	C_a	2.0000	2.0000
Transverse unit load (N/mm)	w_Bt	297.5170	
Lubrication system factor	X_S	1.2000	
Multiple meshing factor	X_mp	1.0000	
Structural factor	X_W	1.0000	1.0000
Structural factor (total)	X_Wtot	1.0000	
Thermal contact factor	B_M	13.7800	13.7800
Multiple path factor	K_mp	1.0000	
Optimal tip relief (μm)	C_eff	10.1300	
Angle factor	X_αβ	0.9680	
Lubricant factor	X_L	0.8790	
Roughness factor	X_R	0.7750	
Mean coefficient of friction	μ_m	0.0450	
Maximum flash temperature (deg)	θ_fl_max	71.9490	
Mean flash temperature (deg)	θ_fl_mean	25.5700	
Total mass temperature (deg)	θ_M	94.4210	
Scuffing temperature (deg)	θ_S	364.8370	
Maximum contact temperature (deg)	θ_Bmax	166.3700	
Safety factor for flash temperature	S_B	3.2980	

3.5. Micro-pitting

Load case 1

	Symbol	Gear1 - Gear2
minimum lubricant film thickness (μm)	h_Ymin	0.1573
minimum specific lubricant film thickness in the contact area	λ_GFmin	0.2621
permissible specific lubricant film thickness	λ_GFP	0.2111
safety factor against micropitting	S_λ	1.2419
transverse tangential load at reference cylinder per mesh (N)	F_t	4609.9890
transverse load in plane of action (N)	F_bt	4948.1230
reduced modulus of elasticity (N/mm ²)	E_r	2.2637×10 ⁵
thermal contact coefficient of pinion (N/(ms ^{0.5} *K))	B_M1	13779.6040
thermal contact coefficient of wheel (N/(ms ^{0.5} *K))	B_M2	13779.6040
failure load stage according to FVA 54/7 (90°)	-	0.0000
effective arithmetic mean roughness value (μm)	R_a	0.6000
roughness factor	X_R	1.1410
lubricant factor	X_L	1.0000
lubrication factor	X_S	1.2000

	Symbol	Gear1 - Gear2
elasticity factor (N/mm ²) ^{0.5}	Z_E	189.8120
helical load factor	K_β	1.3000
load losses factor	H_v	0.1030
mean coefficient of friction	μ_m	0.0530
profile position at min. lubricant film thickness	-	0.0000
tip relief factor	X_Ca	1.0000
load sharing factor	X_A	0.8740
buttressing factor	X_but,A	1.3000
effective tip relief (μm)	C_eff	0.0000
Applied tip relief in calculation (μm)	C_a	0.0000
sum of tangential velocities at point A (m/s)	v_Σ,A	20.8370
sliding velocity at point A (m/s)	v_g,A	-7.3270
tangential velocity on pinion at point A (m/s)	v_r1,A	6.7550
tangential velocity on wheel at point A (m/s)	v_r2,A	14.0820
oil inlet/sump temperature (°C)	θ_oil	80.0000
bulk temperature (°C)	θ_M	97.8310
flash temperature at point A (°C)	θ_fl,A	84.9400
contact temperature at point A (°C)	θ_B,A	182.7710
pressure-viscosity coefficient at 38 °C (m ² /N)	α_38	1.8390×10 ⁻⁸
pressure-viscosity coefficient at bulk temperature (m ² /N)	α_θM	1.3470×10 ⁻⁸
pressure-viscosity coefficient at point A contact temperature (m ² /N)	α_θB,A	8.6990×10 ⁻⁹
kinematic viscosity at 40 °C (mm ² /s)	ν_40	68.0000
kinematic viscosity at 100 °C (mm ² /s)	ν_100	8.8000
kinematic viscosity at point A contact temperature (mm ² /s)	ν_θB,A	2.2890
dynamic viscosity at 38 °C (Ns/m ²)	η_38	0.0650
dynamic viscosity at bulk temperature (Ns/m ²)	η_θM	0.0080
dynamic viscosity at oil inlet/sump temperature (Ns/m ²)	η_θoil	0.0120
dynamic viscosity at point A contact temperature (Ns/m ²)	η_θB,A	0.0020
normal radius of relative curvature at point A (mm)	ρ_n,A	5.6440
material parameter	G_M	3048.8510
local velocity parameter at point A	U_A	6.2470×10 ⁻¹¹
local load parameter at point A	W_A	0.0002
local sliding parameter at point A	S_GF,A	0.1480
local nominal Hertzian contact stress at point A (Method B) (N/mm ²)	p_H,A	887.9020
local Hertzian contact stress at point A including the load factors K (Method B) (N/mm ²)	p_dyn,A	1334.4570

4. Power loss

4.1. Mesh loss

Load case 1

	Symbol	Gear1 - Gear2
Rating Method	-	ISO/TR 14179-2:2001
Input Power (kW)	P	150.0000
Gear ratio	u	2.2414
Tangential speed at working pitch dia. (m/s)	v_{wt}	32.5380
Tangential force at pitch dia. (N)	F_t	4609.9888
Tooth normal force, transverse section (N)	F_{bt}	4948.1229
Arithmetic average roughness of pinion	Ra1	0.6000
Arithmetic average roughness of wheel	Ra2	0.6000
Roughness factor	X_R	0.6000
Sum velocity (m/s)	v_{Σ}	23.6435
Equivalent radius of curvature (mm)	ρ	8.2923
Kinematic viscosity of oil at operating temp. (cSt)	η_{oil}	14.8602
Lubricant type	-	Mineral oil base
Lubricant factor	X_L	1.0000
distributed force on tooth surface(N/m)	F/b	150.0000
Mean coefficient of friction of the gear mesh	μ_{mz}	0.0350
Tooth loss factor	H_v	0.1033
Mesh power loss (W)	P_{VZP}	542.0281
Mesh efficiency (%)	η_z	99.6386

4.2. Churning loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	Changenet (2011)	Changenet (2011)
Rotating speed (rpm)	n	10000.0000	4461.5385
Lubricant	-	Oil: ISO-VG 68	Oil: ISO-VG 68
Gear immersed depth (mm)	h_e	31.0715	69.6431
Oil density (kg/m ³)	ρ	885.0000	885.0000
Operating Viscosity (cSt)	ν	14.8602	14.8602
dynamic viscosity (kg/m.s)	μ	0.0000	0.0000
Reynolds number	Re	76636.3637	87584.4157
Froude number	Fr	3473.3696	1549.6572
Centrifugal acceleration (m/s ²)	γ	14208.3672	3869.7469

	Symbol	Gear1	Gear2
Tip diameter of gear (mm)	d_a	66.1431	143.2862
Pitch diameter of gear (mm)	d_p	62.1431	139.2862
Facewidth (mm)	b	35.0000	40.0000
Dimensionless drag torque	C_m	0.0010	0.0027
Immersion area of the gear (m ²)	S_m	0.0117	0.0373
Total oil churning loss (W)	P	176.8782	1516.9519
Total oil churning torque (N.m)	T	0.1689	3.2468

4.3. Windage loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	None	None
Windage loss (W)	P_WL	0.0000	0.0000
Windage torque (N.m)	T_WL	0.0000	0.0000